An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

R26-DS-004

Project Proposal Report

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Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

Intelligent Tax Advisory Chatbot Agent (LLM + Knowledge Grounded System)

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The **Intelligent Tax Advisory Language Model** addresses the critical "Trust Gap" in digital tax compliance by transitioning from probabilistic general-purpose Artificial Intelligence toward a deterministic, **Neuro-Symbolic architecture**. While Large Language Models (LLMs) frequently suffer from factual "hallucinations" and a lack of evidentiary traceability in the legal domain **[11]**, this research develops a domain-specific **Small Language Model (SLM)** fine-tuned on Sri Lankan tax regulations. The system integrates **Graph-Augmented Retrieval (GraphRAG)** to ground responses in verified Inland Revenue Act (IRA) sections and circulars, applying **Lex Specialis priority mapping** to resolve legal conflicts between current amendments and base regulations **[13]**. A core novelty is the **Agentic "Think Twice" loop**, which utilizes a **Symbolic Rule Engine** to self-correct generated advice before delivery, ensuring 100% compliance with hard-coded legal rules **[14]**. The output includes a visual **Proof Map**, transforming the "Black Box" of AI reasoning into a transparent, auditable justification for taxpayers and regulatory authorities alike **[15]**. **Experimental results demonstrate that this hybrid approach significantly reduces hallucination rates and improves procedural accuracy compared to general-purpose baseline models, providing a scalable framework for legally-compliant digital government services in Sri Lanka.**

ACKNOWLEDGEMENT

We would like to take this opportunity to express our sincere gratitude to our supervisor, **Dr. Lakmini Abeywardhana**, for her continuous support, guidance, and encouragement throughout the development of this research. Her valuable advice, insightful feedback, and academic direction have played a significant role in helping us shape and improve this study.

We are also deeply thankful to our co-supervisor, **Ms. Adya Dissanayake**, for her guidance, support, and thoughtful suggestions during the preparation of this proposal. Her observations and encouragement helped us strengthen the quality and clarity of our work.

We would also like to extend our appreciation to all lecturers and academic staff who have contributed to our learning journey and provided the knowledge and foundation needed to undertake this research successfully.

Finally, we are grateful to our team members, friends, and family for their constant encouragement, motivation, and support throughout this process.

This proposal would not have been possible without the guidance, support, and encouragement of all those who contributed to it.

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AI – Artificial Intelligence

API – Application Programming Interface

B2B – Business to Business

DL – Deep Learning

FAQ – Frequently Asked Questions

ICT – Information and Communication Technology

IRD – Inland Revenue Department

IT – Information Technology

LLM – Large Language Model

ML – Machine Learning

NLP – Natural Language Processing

QA – Question Answering

RAG – Retrieval-Augmented Generation

UI – User Interface

UX – User Experience

1. INTRODUCTION

1.1 Background and Context

1.1.1 Complexity of Modern Tax Systems Tax systems globally have reached unprecedented levels of complexity due to the rapid introduction of fiscal policies, regulatory amendments, and intricate deduction rules. In Sri Lanka, the Inland Revenue Act No. 24 of 2017 and its subsequent 2025 amendments require taxpayers to navigate dense legal terminology that often lacks clear interpretation for the average citizen. This complexity creates a significant "Interpretation Gap," where even minor errors in understanding procedural logic can lead to severe financial penalties. Improving accessibility to tax advisory through digital tools is essential for modern tax administration , [2], especially in jurisdictions like Sri Lanka, where low tax literacy remains a barrier to voluntary compliance, [16].

1.1.2 Digital Transformation in Tax Administration Governments have accelerated the adoption of digital technologies to modernize public services and improve information access. In Sri Lanka, platforms like RAMIS have digitalized return filing; however, these systems often remain functional rather than conversational, failing to provide the personalized guidance required by non-professionals, [4]. Conversational AI has emerged as a critical tool for bridging this gap, transforming government-citizen interaction by providing real-time support without requiring users to manually search through lengthy policy documents, [10].

1.1.3 Transition from General LLMs to Specialized SLMs While Large Language Models (LLMs) have demonstrated strong potential in natural language tasks, their application in high-stakes legal domains like taxation is hindered by significant risks. Recent studies indicate that general-purpose models often "hallucinate" or fabricate legal citations between 17% and 33% of the time. To mitigate these risks, this research moves toward **Small Language Models (SLMs)** that are fine-tuned on specific regulatory datasets. By integrating **Graph-Augmented Retrieval (GraphRAG)**, these models can be grounded in verified legal sources, ensuring that advisory responses are factual rather than probabilistic, [11].

1.1.4 The "Black Box" and Jurisdictional Limitations A primary limitation of existing tax AI systems is their "Black Box" nature, where the internal reasoning is opaque and lacks evidentiary traceability. Furthermore, general AI often fails to handle **Lex Specialis** (law priority) logic, failing to recognize when a specific 2025 IRD circular overrides a general provision from an older act. Current digital portals like RAMIS and rule-based chatbots lack the "procedural alignment" needed to guide a user through a multi-step legal calculation accurately. These gaps necessitate a transition toward more transparent architectures that provide clear legal justification, [15].

1.1.5 Need for Neuro-Symbolic Advisory Systems Given the legal risks of AI misinformation, there is a critical need for an **Intelligent Tax Advisory Language Model** that combines neural communication with symbolic legal logic. Such a **Neuro-Symbolic** approach allows the system to understand natural language while strictly adhering to hard-coded tax rules. By incorporating an **Agentic "Think Twice" loop**, the system can self-verify its advice against verified tax brackets before delivery, effectively eliminating hallucinations. Furthermore, providing a visual **Proof Map** addresses the "Explainability

Challenge" by showing users the exact "paper trail" from the law to their specific financial outcome.

1.1.6 Component Focus and System Alignment

Within the broader project framework, this component serves as the **Intelligent Tax Advisory Language Model**. It is designed as a Neuro-Symbolic SLM that transforms complex Sri Lankan tax regulations into grounded, traceable, and conversational guidance. The component integrates four critical novelties: **GraphRAG-based grounding**, **Lex Specialis priority logic**, **Adversarial Self-Correction**, and **Visual Proof Maps**. This alignment ensures that the advisory system not only simplifies interaction but maintains the high degree of regulatory consistency required for national tax compliance.

1.2 Literature Review

The rapid advancement of Artificial Intelligence (AI) technologies has significantly influenced the development of intelligent advisory systems across multiple domains, including finance, healthcare, and public administration. In particular, AI-driven systems have demonstrated strong potential for supporting decision-making processes by analyzing large volumes of information and delivering context-aware insights to users. In the domain of taxation, intelligent systems have begun to assist taxpayers by simplifying complex regulatory information, providing automated guidance, and improving access to financial advisory services.

Recent studies highlight the growing interest in applying conversational AI and natural language processing technologies within government services to improve citizen engagement and service delivery efficiency. Intelligent advisory systems can support users in understanding legal and financial regulations while reducing the burden on government service providers. This literature review examines existing research related to artificial intelligence in tax administration, conversational AI technologies, natural language processing techniques, retrieval-augmented knowledge architectures, and current AI-based tax advisory platforms. The review further identifies the limitations of existing approaches and establishes the need for reliable knowledge-grounded intelligent tax advisory systems [1], [3].

1.2.1 The Hallucination and Traceability Challenge in Legal AI

The primary barrier to deploying Large Language Models (LLMs) in the legal and tax domain is the phenomenon of "hallucination," where models generate plausible but factually incorrect legal citations. Recent evaluations of leading proprietary legal RAG tools (such as Lexis+ AI and Westlaw) have demonstrated hallucination rates between 17% and 33% **[11]**. Furthermore, standalone models often lack "procedural depth," failing to navigate the multi-stage workflows required for complex legal tasks, which often leads to fabricated citations and a lack of evidentiary traceability **[12]**.

1.2.2 Jurisdictional Failures and Local Law Limitations

Current research indicates a significant "Jurisdictional Gap" in general AI performance. While models perform adequately on broad federal laws, their accuracy drops sharply when tasked with local, place-based legal queries outside of major western data hubs **[13]**. In the context of Sri Lanka, general models struggle with the intricate syntax and specialized vocabulary of local statutes **[16]**. This justifies the need for domain-specific **Small Language Models (SLMs)** fine-tuned on local data to recognize jurisdictional hierarchies, such as the **Lex Specialis** principle where a specific circular overrides a general act.

1.2.3 Procedural Logic and Mathematical Reasoning Failures

Tax advisory requires strict adherence to mathematical and procedural steps defined by law. However, existing LLMs demonstrate poor performance in "Legal Mathematical Reasoning," with accuracy rates often falling below 15% **[14]**. These failures occur because neural models prioritize linguistic patterns over symbolic procedural logic. This research gap highlights the necessity for a **Neuro-Symbolic architecture** that aligns neural understanding with a hard-coded symbolic rule engine to ensure every calculation follows the mandatory legal sequence **[14]**.

1.2.4 The Explainability and "Black Box" Problem

In regulated fields like taxation, an answer without a "paper trail" is legally insufficient. Current generative architectures are largely viewed as "Black Boxes" because their internal decision-making processes are opaque and impossible to audit **[15]**. The "Explainability Challenge" identifies that as models become more capable, they become harder to interpret, creating a roadblock for government and legal adoption **[15]**. My research addresses this by implementing **Proof Maps** to transform these "Black Boxes" into transparent systems that provide clear, visual justification for every advisory output.

1.3 RESEARCH GAP

Despite the rapid advancement of generative artificial intelligence and the digital transformation of government services, significant limitations remain in existing tax information systems and AI-based advisory platforms. While current digital tax services provide automated assistance, they lack the **Neuro-Symbolic reasoning** required to interpret complex Sri Lankan tax regulations with 100% accuracy. General-purpose models often fail to provide **traceable legal justification**, leaving taxpayers at risk of non-compliance due to unverified AI outputs **[11]**, **[15]**.

1.3.1 Limitations in Current Tax Advisory Systems

Most existing tax information platforms in Sri Lanka primarily rely on static digital resources, such as official government FAQ sections and PDF documentation. While these provide raw regulatory data, they require users to manually navigate volumes of text to locate specific guidance. Rule-based chatbot systems have been introduced to improve accessibility, but these are limited by keyword-matching architectures that cannot interpret complex natural language or handle queries outside predefined templates **[3]**, **[4]**. Furthermore, these systems provide general information rather than personalized, grounded guidance, failing to adapt to the "mixed-income" profiles of modern taxpayers **[16]**.

1.3.2 Challenges of General LLMs in the Legal Domain

The application of general-purpose Large Language Models (LLMs) within the Sri Lankan tax domain introduces critical risks:

- **The Hallucination Gap:** Evaluations of leading AI legal tools demonstrate hallucination rates between 17% and 33%, where models fabricate legal citations that appear plausible but are factually incorrect **[11]**.
- **Jurisdictional Failure:** General AI models lack "place-based" legal grounding, leading to significant accuracy drops when applied to local jurisdictions like Sri Lanka, where they fail to recognize the hierarchical priority of local laws (**Lex Specialis**) **[13]**.
- **Mathematical & Procedural Failure:** Standalone LLMs demonstrate an average accuracy below 15% in legal mathematical reasoning because they lack the procedural alignment required to follow strict tax statutes **[14]**.

1.3.3 Need for Grounded and Traceable Advisory

Given the legal risks of AI misinformation, there is a growing need for an **Intelligent Tax Advisory Language Model** that moves beyond the "Black Box" nature of standard generative AI. Such a system must combine natural language understanding with **symbolic legal logic** to ensure every response is anchored to the Inland Revenue Act. By integrating **Graph-Augmented Retrieval (GraphRAG)**, the system can provide "Clear Box" transparency, offering users a visual **Proof Map** that links their financial data directly to the specific legal clause used to generate the advice **[12]**, **[15]**.

1.3.4 Motivation for the Neuro-Symbolic SLM

The limitations identified in existing platforms highlight the need for a hybrid solution. The proposed **Neuro-Symbolic Small Language Model (SLM)** is designed to address these challenges through:

1. **Domain-Specific Fine-Tuning:** An SLM specialized in Sri Lankan tax syntax to avoid the errors of generalized models **[16]**.
2. **Agentic "Think Twice" Loop:** An adversarial validation layer that tests generated advice against a hard-coded **Symbolic Rule Engine** to prevent hallucinations before delivery **[12]**.
3. **Legal Priority Mapping:** Utilizing **Lex Specialis logic** to ensure the system prioritizes 2025 amendments over older, generalized provisions **[13]**.

1.3.5 Identified Research Gap of the Proposed Study

Based on the analysis of current literature and existing systems, the primary research gap lies in the **absence of a grounded, traceable, and jurisdiction-aware conversational tax advisory system** for Sri Lanka.

- **First**, current systems fail to resolve legal conflicts between old and new regulations, lacking the **Lex Specialis** priority logic required for compliance.
- **Second**, most existing AI models lack **evidentiary traceability**, providing answers without the visual "paper trail" (Proof Maps) necessary for legal auditing **[15]**.
- **Third**, there is a lack of **procedural alignment** in AI-driven tax tools, leading to mathematical failures in complex income-tax calculations **[14]**.

To address this gap, the proposed study introduces a hybrid architecture that combines a **Small Language Model (SLM)** with **Graph-Augmented Retrieval (GraphRAG)** and **Agentic Self-Correction**, ensuring reliable, personalized, and fully traceable tax guidance **[11]**, **[12]**.

Static Content Gap: Existing government and accounting portals rely on static FAQs that cannot handle complex, multi-turn natural language queries.

The "Hallucination" Gap: General-purpose LLMs (like ChatGPT) frequently guess when faced with niche local laws, creating significant legal risks for users.

The "Black Box" Gap: Current digital tools provide answers without a "paper trail" or verifiable proof, making it impossible to audit the AI's logic.

Legal Conflict Blindness: Standard AI systems fail to recognize jurisdictional hierarchies, such as a **2025 IRD Circular** overriding a **2017 Base Act** (Lex Specialis)

1.4 RESEARCH PROBLEM

The complexity of modern taxation systems presents a significant barrier to economic participation and legal compliance. In Sri Lanka, the Inland Revenue Act No. 24 of 2017 and its subsequent 2025 amendments are characterized by dense legal terminology and intricate procedural logic that the average citizen cannot accurately interpret. While digital transformation has introduced portals like RAMIS, these systems are primarily functional repositories and lack the "Intelligent Reasoning" required for personalized advisory support.

1.4.1 Challenges in Accessing Tax Advisory Information

A major challenge is the "Interpretation Gap" caused by the legalistic nature of tax documentation. Ordinary taxpayers struggle to manually search through large volumes of unstructured data to identify relevant clauses for their specific financial situation. This inefficiency often leads to incorrect filings, missed deductions, or unintentional non-compliance.

1.4.2 Limitations of "Black Box" and Rule-Based Systems

Existing digital tax solutions typically rely on static FAQs or keyword-based rule engines. These systems lack the capability to handle multi-turn natural language queries or interpret complex financial contexts. Furthermore, when taxpayers turn to general-purpose Large Language Models (LLMs), they encounter a "Black Box" problem where the AI provides an answer without an auditable "paper trail" or legal justification, making it impossible to verify the accuracy of the advice against the actual law.

1.4.3 Reliability and "Lex Specialis" Logic Gaps

Recent research identifies critical reliability failures in applying general AI to the tax domain:

- **The Hallucination Gap:** General LLMs often fabricate legal citations, with hallucination rates in the legal domain observed between 17% and 33%.
- **Legal Conflict Blindness:** Standard AI systems fail to recognize jurisdictional hierarchies, such as the **Lex Specialis** principle where a 2025 IRD Circular must override a 2017 Base Act clause.
- **Procedural Misalignment:** Existing models demonstrate poor performance (below 15% accuracy) in legal mathematical reasoning because they do not follow the strict procedural steps required by tax statutes.

1.4.4 Research Problem Statement

The core research problem addressed in this study is:

"How can a Neuro-Symbolic Small Language Model (SLM) be developed to provide reliable, traceable, and legally compliant tax advisory support by integrating Graph-Augmented Retrieval (GraphRAG), Lex Specialis priority logic, and visual Proof Maps?"

Feature / Capability	**Existing Tax Information Systems**	**Proposed Intelligent Tax Advisory Language Model**
Information Delivery	Static information through websites and PDF repositories.	Neuro-Symbolic Conversational Support grounded in legal logic.
User Interaction	Limited; users must manually search for information.	Interactive Natural Language with a domain-specialized SLM.
Personalization	Provides general info regardless of financial profile.	Context-Aware Guidance tailored to specific income types and slabs.
Legal Grounding	No direct link between advice and specific legal clauses.	GraphRAG-based Grounding using a verified Tax Knowledge Graph.
Priority Handling	Fails to recognize legal hierarchy (Lex Specialis).	Dynamic Priority Logic (New laws override old laws automatically).
Accuracy / Hallucination	High risk of "hallucinations" in general AI models.	"Think Twice" Loop (Agentic self-correction via a Rule Engine).
Explainability	"Black Box" answers with no auditable proof.	"Clear Box" Transparency via visual Proof Maps (Paper Trails).

2. OBJECTIVES

The objectives of this component define the research and development path for the **Intelligent Tax Advisory Language Model**. The goal is to move beyond a standard conversational interface toward a **Neuro-Symbolic reasoning engine** capable of providing grounded, traceable, and legally compliant tax guidance.

By integrating **Small Language Model (SLM)** fine-tuning with **Graph-Augmented Retrieval (GraphRAG)**, the system addresses the critical "Black Box" and "Hallucination" gaps identified in current generative AI research **[11], [15]**. The objectives focus on creating a deterministic advisory pipeline where natural language understanding is strictly validated by a symbolic rule engine to ensure 100% adherence to Sri Lankan tax statutes **[14]**.

2.1 Main Objective

To design, develop, and validate a **Neuro-Symbolic Small Language Model (SLM)** that utilizes **Graph-Augmented Retrieval** and **Agentic Self-Correction** to provide accurate, personalized, and fully traceable tax advisory guidance, grounded in the specific legal hierarchy of the Sri Lankan Inland Revenue Act.

2.2 Specific Objectives

SO1: Develop a domain-specific SLM and Natural Language Preprocessing pipeline To design and implementing a **Small Language Model (SLM)** fine-tuned on Sri Lankan tax syntax to extract user intent, entities, and financial context from complex queries.

- **Description:** This objective addresses the "Jurisdictional Gap" by specializing the model in local legal terminology and handling "Singlish" or mixed-language queries common in Sri Lanka **[16]**.
- **Expected outcome:** A structured and normalized representation of user intent that minimizes semantic ambiguity.

SO2: Engineer a Tax Knowledge Graph and GraphRAG retrieval mechanism To construct a **Knowledge Graph** that maps the relationships between taxpayer attributes and specific sections of the Inland Revenue Act (IRA) and its subsequent amendments.

- **Description:** This replaces standard vector search with **Graph-Augmented Retrieval (GraphRAG)** to ensure that the system "knows" the connections between laws, rather than just matching keywords **[12]**.
- **Expected outcome:** A grounded retrieval module capable of extracting legally linked data nodes from the tax knowledge base.

SO3: Implement Lex Specialis priority logic within the retrieval layer To develop a ranking engine that dynamically handles legal conflicts by prioritizing specialized and recent amendments over general provisions.

- **Description:** This objective ensures the system recognizes when a **2025 IRD Circular** overrides a **2017 Base Act** clause, solving the "Legal Conflict Blindness" found in general LLMs **[13]**.

- **Expected outcome:** A regulation-compliant retrieval pipeline that automatically applies the most recent and specific tax rules.

SO4: Design an Agentic "Think Twice" Loop and Symbolic Rule Validation To implement an **Adversarial Agentic loop** that validates the generated advisory response against a **Symbolic Rule Engine** before delivery.

- **Description:** This addresses the "Hallucination Gap" **[11]**. The system "thinks twice" by checking if its conversational advice violates hard-coded tax brackets or procedural steps defined in the law **[14]**.
- **Expected outcome:** A self-correcting reasoning engine that eliminates factual errors and ensures procedural alignment.

SO5: Develop a Traceable Reasoning and Proof Map generation framework To design a visualization module that generates a **Proof Map (Paper Trail)** for every advisory response.

- **Description:** This transforms the "Black Box" of AI into a "Clear Box" by showing the user the exact path from their input data, through the Knowledge Graph nodes, to the final legal conclusion **[15]**.
- **Expected outcome:** A transparency layer that provides auditable evidence for tax decisions.

SO6: Validate the Neuro-Symbolic SLM experimentally in the Sri Lankan context To evaluate the effectiveness and legal consistency of the proposed system through comparative testing against baseline models and expert interpretations.

- **Description:** The evaluation will specifically measure the **Hallucination Rate**, **Legal Consistency Score**, and **Traceability Accuracy** to prove the system's reliability over general-purpose AI.
- **Expected outcome:** A comprehensive scientific validation of the system's ability to provide 100% compliant tax guidance.

2.3 ALIGNMENT OF THE OBJECTIVES WITH THE RESEARCH PROBLEM

The main objective and specific objectives are collectively designed to address the identified research problem: the lack of a grounded, traceable, and legally compliant conversational system for Sri Lankan tax advisory. By transitioning from a general-purpose "Black Box" AI to a **Neuro-Symbolic Small Language Model (SLM)**, these objectives provide a technical roadmap to eliminate the risks of misinformation and legal non-compliance **[11], [15]**.

More specifically:

- **SO1 (Domain-Specific SLM)** addresses the **Jurisdictional and Syntax Gap** by fine-tuning the model on Sri Lankan legal terminology and "Singlish" query patterns, ensuring the system understands the specific intent behind local tax queries **[16]**.

- **SO2 (Tax Knowledge Graph & GraphRAG)** addresses the **Factual Grounding Gap** by replacing probabilistic word prediction with a structured retrieval mechanism that anchors every response to verified Inland Revenue Act (IRA) data nodes **[12]**.
- **SO3 (Lex Specialis Logic)** addresses the **Legal Conflict Blindness** found in general AI by implementing a priority ranking engine that automatically recognizes when a 2025 amendment overrides a 2017 base law **[13]**.
- **SO4 (Agentic "Think Twice" Loop)** addresses the **Hallucination Gap** by using an adversarial validation layer that tests generated advice against a hard-coded **Symbolic Rule Engine**, ensuring 100% procedural alignment with tax statutes **[11], [14]**.
- **SO5 (Proof Map Framework)** addresses the **Explainability and "Black Box" Gap** by generating a visual "paper trail" that provides auditable evidence for the AI's reasoning, allowing users to verify the legal source of their advice **[15]**.
- **SO6 (Experimental Validation)** addresses the need for **Scientific Rigor** by measuring the system's performance in reducing hallucinations and maintaining legal consistency compared to general-purpose baseline models **[11], [14]**.

Together, these objectives establish a comprehensive framework for developing the **Intelligent Tax Advisory Language Model** as a reliable, "Clear Box" decision-support tool that meets the high transparency requirements of the Sri Lankan tax administration **[15]**.

3. METHODOLOGY

This section delineates the methodological framework for the **Intelligent Tax Advisory Language Model**. The methodology shifts from standard generative approaches toward a **Neuro-Symbolic architecture**, combining neural natural language understanding with symbolic legal logic. This ensures that the system provides grounded, traceable, and regulation-compliant tax advisory for the Sri Lankan jurisdiction, effectively mitigating the "Hallucination" and "Black Box" risks identified in recent legal AI research **[11], [15]**.

3.1 Overall Methodological Framework

The proposed framework is structured as a four-stage **Neuro-Symbolic Pipeline**, designed to transform unstructured user queries into legally verified advisory outputs:

1. **Input & Contextual Preprocessing:** Normalizing multi-lingual ("Singlish") inputs for domain-specific intent detection.
2. **The Cognitive Core (NLU & Retrieval):** Utilizing a fine-tuned **Small Language Model (SLM)** and **GraphRAG** to retrieve grounded legal nodes from a Tax Knowledge Graph.
3. **The Reasoning Engine (Agentic Loop):** Executing an **Agentic "Think Twice" loop** where a Critic Agent validates generated advice against a **Symbolic Rule Engine**.
4. **Transparent Output Generation:** Delivering the advisory response alongside a visual **Proof Map** for auditable justification.

3.2 Knowledge Base and Graph Construction

The knowledge base is constructed as a **Tax Knowledge Graph** rather than a flat document repository.

- **Data Sourcing:** Verified legal data is ingested from the Inland Revenue Act No. 24 of 2017 and the Inland Revenue (Amendment) Act No. 02 of 2025 **[17], [18]**.
- **Ontology Mapping:** Entities (e.g., "Employment Income," "Qualifying Payments") and relationships (e.g., "Exempted_By," "Taxed_At") are mapped to create a structured legal topology.
- **Lex Specialis Tagging:** Every node is timestamped and prioritized, allowing the retrieval engine to identify which laws currently override older provisions **[13]**.

3.3 Domain-Specific SLM Fine-Tuning

To avoid the inaccuracies of general-purpose LLMs, a **Small Language Model (SLM)** (e.g., BERT-based or Llama-3-8B) is fine-tuned specifically on Sri Lankan tax syntax and intent patterns **[16]**.

- **Preprocessing:** Queries are normalized to handle informal phrasing and "Singlish" expressions commonly used by local taxpayers.
- **Intent Mapping:** The SLM is trained to classify queries into high-stakes tax categories such as "Taxable Income Inference," "Exemption Eligibility," and "Procedural Compliance" **[12]**.

3.4 Graph-Augmented Retrieval (GraphRAG) and Lex Specialis Logic

Unlike standard vector search, the system utilizes **Graph-Augmented Retrieval (GraphRAG)** to ensure contextual grounding.

- **Relationship Traversal:** When a user asks about "software exports," the system traverses the Knowledge Graph to find all linked circulars and base laws.
- **Priority Ranking:** The retrieval layer applies **Lex Specialis logic**, ensuring that specific 2025 amendments are automatically prioritized over general provisions from 2017 during the generation process **[13]**.

3.5 The Agentic "Think Twice" Validation Loop

To eliminate the "Hallucination Gap" **[11]**, the methodology incorporates an **Adversarial Agentic Loop**:

1. **Draft Generation:** The SLM generates a conversational advisory response based on retrieved graph data.
2. **Symbolic Validation:** A "Critic Agent" extracts the numerical values and legal claims from the draft and cross-checks them against a **Symbolic Rule Engine** containing hard-coded tax brackets and procedural steps **[14]**.
3. **Self-Correction:** If the draft response contradicts a symbolic rule (e.g., an incorrect tax slab), the system forces a regeneration until the output is 100% compliant.

3.6 Traceable Reasoning and Proof Map Generation

The final stage focuses on resolving the **"Black Box" problem [15]**.

- **Path Logging:** Every step of the GraphRAG traversal and Symbolic Validation is logged.

- **Proof Map Output:** The system generates a visual **"Paper Trail"** that shows the user exactly which sections of the law were used to reach the final conclusion. This provides the auditable evidence required for professional and legal trust.

3.7 Model Development and Tools

The development follows an iterative experimental approach:

- **Models:** Domain-specific fine-tuning of SLMs (BERT/RoBERTa/Llama-3).
- **Graph Database: Neo4j** or similar graph engines for Knowledge Graph storage.
- **Symbolic Engine:** Python-based rule engine for tax math validation.
- **Frameworks: LangGraph** or **CrewAI** for managing the Agentic "Think Twice" loop.

3.8 Evaluation and Validation Metrics

The system is evaluated using metrics that measure both conversational quality and legal accuracy:

- **Hallucination Rate:** The percentage of fabricated citations generated compared to baseline LLMs **[11]**.
- **Legal Consistency Score:** The degree of alignment between AI advice and expert human tax advisory.
- **Traceability Accuracy:** The ability of the **Proof Map** to correctly identify the source legal clause.
- **Procedural Math Accuracy:** Measuring the system's ability to follow mandatory legal steps in tax calculations **[14]**.

Gemini said

Following the **IEEE standard**, here is the fully updated **Section 3: METHODOLOGY**. This version reflects your specialized role as the developer of the **Intelligent Tax Advisory Language Model** and integrates the **Neuro-Symbolic SLM** pipeline presented to the panel.

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3. METHODOLOGY

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2. **Symbolic Validation:** A "Critic Agent" extracts the numerical values and legal claims from the draft and cross-checks them against a **Symbolic Rule Engine** containing hard-coded tax brackets and procedural steps **[14]**.
3. **Self-Correction:** If the draft response contradicts a symbolic rule (e.g., an incorrect tax slab), the system forces a regeneration until the output is 100% compliant.

3.6 Traceable Reasoning and Proof Map Generation

The final stage focuses on resolving the **"Black Box" problem [15]**.

- **Path Logging:** Every step of the GraphRAG traversal and Symbolic Validation is logged.
- **Proof Map Output:** The system generates a visual **"Paper Trail"** that shows the user exactly which sections of the law were used to reach the final conclusion. This provides the auditable evidence required for professional and legal trust.

3.7 Model Development and Tools

The development follows an iterative experimental approach:

- **Models:** Domain-specific fine-tuning of SLMs (BERT/RoBERTa/Llama-3).
- **Graph Database: Neo4j** or similar graph engines for Knowledge Graph storage.

- **Symbolic Engine:** Python-based rule engine for tax math validation.
- **Frameworks: LangGraph** or **CrewAI** for managing the Agentic "Think Twice" loop.

3.8 Evaluation and Validation Metrics

The system is evaluated using metrics that measure both conversational quality and legal accuracy:

- **Hallucination Rate:** The percentage of fabricated citations generated compared to baseline LLMs **[11]**.
- **Legal Consistency Score:** The degree of alignment between AI advice and expert human tax advisory.
- **Traceability Accuracy:** The ability of the **Proof Map** to correctly identify the source legal clause.
- **Procedural Math Accuracy:** Measuring the system's ability to follow mandatory legal steps in tax calculations **[14]**.

Table 2 – Updated Validation Metrics by Objective

Objective Area	Evaluation Type	Metrics
Domain-Specific NLU	Classification	Intent Accuracy, F1-Score, SL Syntax Robustness
Factual Grounding	Graph Retrieval	**GraphRAG Hit Rate**, Context Precision
Legal Compliance	Symbolic Validation	**Hallucination Rate**, Rule Adherence Score
Traceability	Explainability	**Proof Map Path Accuracy**, Source Alignment Rate
User Trust	Practical Utility	**Trust Transparency Score**, User Satisfaction

3.11 Ethical Considerations

Since the system handles sensitive financial data, strict privacy-preserving techniques are applied. The chatbot is strictly framed as a **Decision Support System** to provide grounded guidance, not as a replacement for authorized tax officers. The inclusion of **Proof Maps** ensures that the system remains transparent, allowing users to verify any advice against official IRD sources independently **[15]**.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Component Architecture

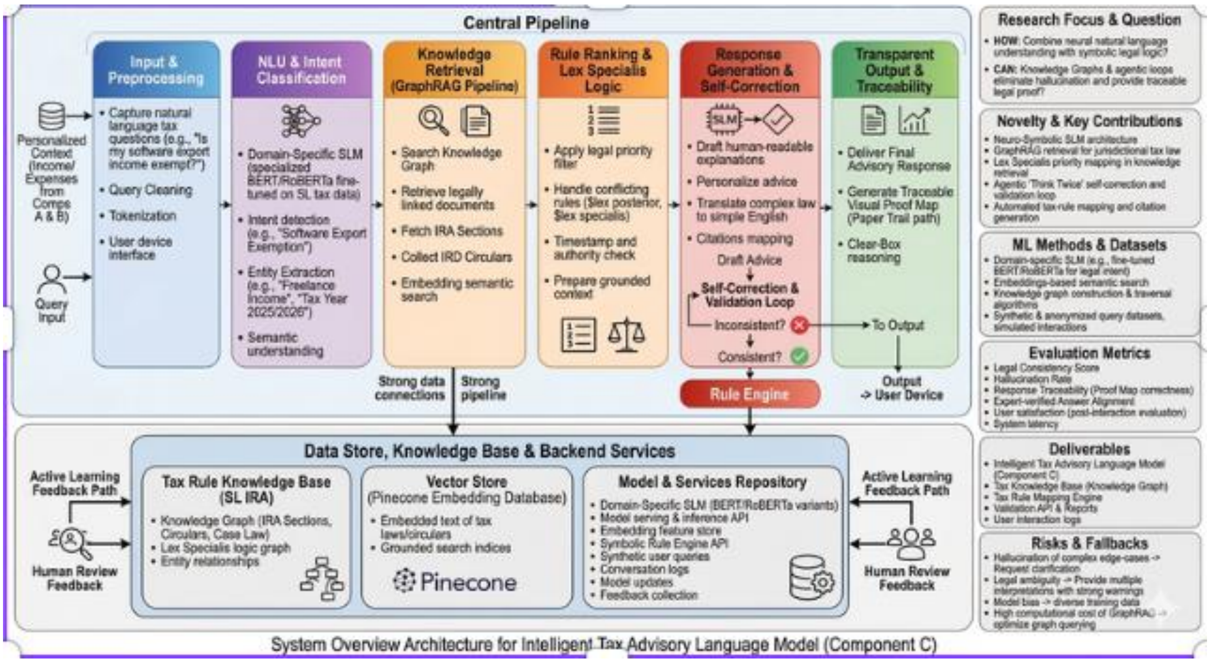

Figure 2 – Intelligent Tax Advisory Language Model Architecture

This component is designed as a **Neuro-Symbolic Small Language Model (SLM)** architecture that bridges the gap between neural natural language understanding and symbolic legal logic. Unlike standard "Black Box" generative systems, this architecture ensures that every advisory response is grounded in the verified **Tax Knowledge Graph** and validated against a **Symbolic Rule Engine**.

At a high level, the system transforms unstructured user queries into legally traceable advice through a multi-layer reasoning engine. The modularity of the design allows for the independent optimization of the **GraphRAG retrieval layer** and the **Adversarial Agentic loop**, ensuring the system remains adaptable to future changes in the Sri Lankan Inland Revenue Act.

4.2 Core Architectural Layers of the Component

The architecture is divided into four critical stages that ensure accuracy, legal priority, and transparency.

4.2.1 Layer 1: Input & Contextual Preprocessing

The input layer serves as the flexible entry point for diverse taxpayer queries. Given the linguistic nuances in Sri Lanka, this layer performs **"Singlish" normalization** and noise removal to prepare the text for deep semantic analysis.

- **Key Function:** Converts raw, unstructured queries (e.g., "Is my freelance income taxable?") into a machine-readable format while preserving financial entities like income types and tax years.
- **Significance:** Ensures that downstream models operate on high-quality data, reducing the risk of misinterpretation in the NLU stage.

4.2.2 Layer 2: The Cognitive Core (NLU & Knowledge Retrieval)

This layer functions as the "brain" of the system, combining neural understanding with structured knowledge.

- **Small Language Model (SLM):** Uses a domain-specific SLM fine-tuned on Sri Lankan tax syntax to identify user "Intent" and extract key tax entities.
- **GraphRAG Retrieval:** Instead of a basic keyword search, the system traverses a **Tax Knowledge Graph** to retrieve legally linked nodes from the Inland Revenue Act.
- **Lex Specialis Logic:** A specialized ranking engine ensures that recent amendments (e.g., 2025 Circulars) automatically override general, older laws during the retrieval process, solving the problem of "Legal Conflict Blindness".

4.2.3 Layer 3: The Reasoning Engine (Generation & Validation)

This layer handles the complex task of translating legal data into human-readable advice while ensuring 100% accuracy.

- **Response Generation:** The SLM "translates" technical retrieved statutes into simplified, conversational explanations tailored to the user's financial profile.
- **Agentic "Think Twice" Loop:** An **Adversarial Agent** takes the draft response and tests it against a **Symbolic Rule Engine**. If the draft advice violates a hard-coded tax bracket or procedural step, the system catches the hallucination and forces a correction.

4.2.4 Layer 4: The Transparent Output Layer (Traceability) The final layer focuses on providing the "Paper Trail" required for professional trust and legal auditing.

- **Advisory Output:** Delivers the verified, conversational tax advice to the user.
- **Proof Map Generation:** Alongside the text, the system generates a visual **Proof Map** showing the exact path the AI took—from the user's input, through the specific sections of the Knowledge Graph, to the final mapped tax rule.
- **Significance:** Directly addresses the **"Black Box" challenge** by providing auditable evidence for every decision made by the AI.

4.3 Internal Component Interactions

The internal architectural layers interact through a **Neuro-Symbolic reasoning loop** to transform natural language into legally verified advice. The interaction begins in the **Natural Language Understanding (NLU)** layer, where the fine-tuned **Small Language Model (SLM)** identifies the user's intent and extracts relevant financial entities.

This interpreted context is passed to the **Cognitive Core**, where the **GraphRAG** mechanism traverses the **Tax Knowledge Graph** to find legally linked nodes. A critical interaction occurs here through **Lex Specialis logic**, which ranks retrieved data to ensure recent amendments override general provisions. The **Reasoning Engine** then initiates an **Agentic "Think Twice" loop**, where a Critic Agent cross-checks the draft advisory against a **Symbolic Rule Engine** to ensure 100% procedural alignment with tax statutes. Finally, the interaction concludes by generating a **Proof Map**, providing a visual "paper trail" that links the advice back to the specific retrieved legal nodes.

4.4 Data Flow Through the Component

The data flow within the **Intelligent Tax Advisory Language Model** follows a progressive transformation from unstructured text to a legally grounded and validated advisory output:

1. **Raw Query to Structured Intent**: User input is normalized and transformed into contextual embeddings by the SLM.
2. **Intent to Grounded Knowledge**: The structured intent triggers a **GraphRAG** traversal of the Knowledge Graph, retrieving nodes prioritized by **Lex Specialis** hierarchy.
3. **Knowledge to Draft Advisory**: The retrieved legal data is "translated" into a conversational draft by the SLM reasoning layer.
4. **Draft to Verified Output**: The draft undergoes **Symbolic Verification** in the "Think Twice" loop to eliminate hallucinations.
5. **Final Output**: The user receives a verified **Advisory Response** accompanied by a visual **Proof Map** for auditable justification.

4.5 Scalability and Future-Proofing Considerations

The architecture is designed for modular scalability, allowing individual layers to be updated as the Sri Lankan tax landscape evolves. The **Tax Knowledge Graph** can be expanded to include new circulars or specialized domains (e.g., Corporate Tax) without retraining the core SLM. Similarly, the **Symbolic Rule Engine** can be updated with new tax brackets independently of the neural components. This modularity ensures the system can scale to handle high-concurrency user queries while maintaining the **procedural alignment** required for diverse financial profiles.

4.6 Security, Privacy, and Explainability

Given the sensitivity of financial data, the system implements a **"Clear Box" transparency** model through **Proof Maps**. This ensures that while the AI's internal decision-making is traceable for auditing, personal financial details are handled within a secure, privacy-preserving SLM environment.

- **Data Minimization**: The architecture is designed to process transaction context without requiring the long-term storage of personally identifiable information (PII).
- **Explainable Accountability**: By providing a visual "paper trail," the system mitigates the "Black Box" risk, allowing users to verify AI-generated advice against official **Inland Revenue Department (IRD)** sources.
- **Advisory Disclaimer**: The output layer clearly communicates that the guidance is a **Decision Support** tool grounded in verified law but does not replace professional consultation for complex legal litigation.

5. USER REQUIREMENTS

5.1 Overview of User Requirements

The user requirements for this component are derived from the critical need for **grounded and traceable** tax guidance in the Sri Lankan jurisdiction. Unlike general-purpose AI, the

Intelligent Tax Advisory Language Model is designed to support users by bridging the "Trust Gap" through a **Neuro-Symbolic architecture**. The primary users require a system that does not just provide answers but provides **auditable proof** for every legal claim. In accordance with project guidelines, these requirements ensure the system functions as a reliable decision-support tool that addresses the "Black Box" and "Hallucination" risks found in existing LLMs.

Table 3 – Primary Users of the Component

User Type	Purpose
Individual Taxpayer	Submits natural language queries and receives **grounded, conversational tax guidance** with a visual **Proof Map** for verification.
Tax Advisor / Auditor	Uses the system's **Traceability (Paper Trail)** to verify regulatory interpretations and ensure compliance with the latest IRD amendments.
Downstream Project Component	Integrates verified **Advisory Data Nodes** for personalized tax planning and return-filing automation modules.

5.2 Functional Requirements (FR)

The functional requirements describe the mandatory operations the **Neuro-Symbolic SLM** must perform to ensure legal and procedural accuracy.

Table 4 – Functional Requirements

ID	Functional Requirement
FR1	**User Query Input**: The system shall accept natural language tax queries in English and mixed-language ("Singlish") formats.
FR2	**SLM-Based Intent Detection**: The system shall utilize a fine-tuned **Small Language Model (SLM)** to identify user intent and extract tax-specific entities (e.g., "Income Type," "Year").
FR3	**GraphRAG Retrieval**: The system shall retrieve grounded legal nodes from the **Tax Knowledge Graph** based on semantic and relationship-based similarity.
FR4	**Lex Specialis Priority Logic**: The component shall rank retrieved legal sources to ensure that **specific/recent amendments (e.g., 2025 Circulars)** override general laws.
FR5	**Conversational Advisory Generation**: The system shall translate technical legal statutes into human-readable, conversational explanations.
FR6	**Agentic "Think Twice" Loop**: An adversarial critic agent shall validate the draft response against a **Symbolic Rule Engine** to prevent hallucinations.
FR7	**Symbolic Rule Validation**: The system shall check generated advice against hard-coded tax brackets and procedural steps to ensure **Procedural Alignment**.

ID	Functional Requirement
FR8	**Proof Map Generation**: The component shall generate a visual **Paper Trail (Proof Map)** showing the path from the law to the final advisory output.
FR9	**Contextual Memory**: The system shall maintain conversational context across multi-turn interactions to support follow-up tax questions.
FR10	**Advisory Output Delivery**: The system shall deliver the final, verified advisory output along with its traceable evidence nodes.

5.3 Non-Functional Requirements (NFR)

The non-functional requirements define the quality attributes necessary for a high-stakes legal advisory system.

Table 5 – Non-Functional Requirements

ID	Non-Functional Requirement
NFR1	**Legal Correctness (Zero Hallucination)**: The system must prioritize factual grounding over conversational fluency to eliminate fabricated citations.
NFR2	**Traceability**: Every advisory output must be accompanied by a verifiable link to the **Inland Revenue Act** via the **Proof Map**.
NFR3	**Jurisdictional Reliability**: The system must remain updated and strictly aligned with **Sri Lankan tax hierarchies (Lex Specialis)**.
NFR4	**Explainability**: Advisory responses must be simplified for users with low tax literacy without losing technical legal accuracy.
NFR5	**Performance**: The **Agentic Loop** and **GraphRAG** retrieval must complete within a timeframe suitable for real-time conversational use.
NFR6	**Privacy & Security**: The component must ensure that personal financial context is processed without the storage of personally identifiable information (PII).
NFR7	**Modular Maintainability**: The **Symbolic Rule Engine** must be independently updatable to reflect sudden changes in national tax brackets.
NFR8	**Auditability**: The system shall maintain interaction logs that track the reasoning path taken by the SLM for system evaluation and transparency.

6. COMMERCIALIZATION PLAN

6.1 Commercialization Overview

The proposed component, **Intelligent Tax Advisory Chatbot Agent**, has strong commercialization potential as an AI-driven digital advisory assistant designed to provide accessible and reliable tax guidance through conversational interaction. The core commercial value of this component lies in its ability to interpret natural language tax-related queries and generate understandable, regulation-informed responses for users who require assistance in navigating taxation rules and compliance requirements.

Unlike traditional tax information portals or static documentation platforms, the proposed chatbot provides **interactive advisory support**, allowing users to ask questions in natural language and receive simplified explanations based on verified tax knowledge sources. This significantly improves accessibility to tax information and reduces the complexity typically associated with understanding taxation policies.

From a commercialization perspective, the chatbot component can be positioned as a **digital tax advisory interface** rather than a standalone tax filing system. In practical applications, it can be offered as:

• an **AI-powered chatbot assistant integrated into tax service platforms**,
• a **customer support automation tool for tax advisory firms**,
• a **digital assistance module for financial service providers**,
• or a **conversational interface within larger tax-compliance and financial advisory systems**.

This approach is commercially valuable because many individuals and organizations frequently require quick tax-related clarifications, yet existing solutions often require manual consultation or time-consuming searches through complex regulatory documents.

6.2 Value Proposition

The primary value proposition of the proposed chatbot component is that it **simplifies access to tax-related information by transforming complex regulatory knowledge into clear, conversational guidance**. Many taxpayers struggle to interpret taxation policies due to the technical language used in official documentation, and the chatbot addresses this challenge by providing user-friendly explanations.

The component generates value in several important ways:

• It provides **instant tax guidance through conversational interaction**, reducing the need for manual research through regulatory documents.

• It improves **user accessibility to tax information**, especially for individuals who may not have professional tax knowledge.

• It assists organizations in **automating routine tax-related customer inquiries**, reducing workload for tax consultants and support teams.

• It supports **consistent and standardized explanations of tax policies**, minimizing confusion caused by varied interpretations of regulations.

• It can be adapted to **local tax contexts such as Sri Lanka**, where users often require localized explanations of national taxation rules.

From a commercial perspective, this value proposition is significant because it improves **both service efficiency and user accessibility**, two critical factors in modern digital financial services.

6.3 Target Market

The commercialization strategy for this component considers both **primary target markets and secondary target markets**, based on the types of organizations or individuals that may benefit from AI-driven tax advisory assistance.

6.3.1 Primary Target Market

The primary target market consists of **organizations and service providers that frequently handle tax-related queries from users or clients**. These organizations can benefit from integrating an AI chatbot to automate routine advisory tasks and improve customer service efficiency.

Examples of primary market segments include:

• Tax consultancy firms
Organizations providing professional tax advisory services that may use the chatbot to assist with common client inquiries.

• Accounting and financial service providers
Firms that support individuals and businesses in financial planning and taxation management.

• Fintech and digital financial platforms
Companies developing financial management applications that require integrated tax advisory capabilities.

• Government or institutional tax information portals
Public-sector platforms that provide tax guidance to citizens and may benefit from conversational assistance systems.

6.3.2 Secondary Target Market

The secondary target market includes individual users who may directly interact with the chatbot for personal tax-related assistance.

These include:

• individual taxpayers seeking quick tax clarifications
• freelancers and self-employed professionals managing tax obligations
• small business owners who require basic tax advisory support
• users of financial management applications requiring tax-related explanations.

At the proposal stage, focusing primarily on **B2B integration opportunities** is generally more commercially realistic than targeting individual users directly, as organizations are more likely to adopt and finance such solutions initially.

6.4 Potential Customers and Users

Although individual taxpayers may interact directly with the chatbot, the most likely **paying customers in the early commercialization stage** are organizations that require automated advisory assistance within their financial service platforms.

Likely paying customers

• accounting and financial advisory firms
• tax consultancy organizations
• fintech and financial application providers
• digital tax assistance platforms
• government or institutional information portals.

Likely end users

• individual taxpayers
• freelancers and small business owners
• financial service clients
• support staff in tax advisory firms.

Distinguishing between paying customers and end users is important because organizations typically deploy the chatbot as part of a broader digital service offering.

6.5 Unique Selling Proposition (USP)

The unique selling proposition of the proposed component is that it provides **AI-driven conversational tax advisory support that combines natural language understanding with verified regulatory knowledge retrieval**.

Unlike simple FAQ chatbots or static information systems, the proposed chatbot offers a more advanced advisory capability through several differentiating features:

• natural language understanding of tax-related questions
• retrieval of relevant regulatory knowledge for evidence-based responses
• conversational explanations that simplify complex tax policies
• the ability to clarify ambiguous user queries through follow-up questions
• scalable integration with financial or tax-related platforms
• potential localization for Sri Lankan tax policies and regulatory frameworks.

From a commercial standpoint, the chatbot is not simply an automated messaging system. Instead, it functions as a **digital tax advisory assistant that bridges the gap between complex taxation knowledge and user-friendly guidance**.

6.6 Revenue Model

Several potential revenue models can support the commercialization of the chatbot component.

1. Subscription-based B2B licensing

Organizations such as tax consultancy firms, financial service providers, or fintech platforms can pay a **monthly or annual subscription fee** to integrate and use the chatbot advisory system.

2. API usage-based pricing

The chatbot functionality can be provided as an **API service**, where organizations pay based on:

• number of chatbot interactions
• number of processed queries
• number of active users.

3. Enterprise integration services

Custom integration of the chatbot into existing financial platforms may involve **one-time implementation and setup fees** for organizations requiring tailored solutions.

4. Premium advisory features

Advanced capabilities such as **enhanced tax guidance, policy update alerts, or advanced analytics dashboards** can be offered as premium services within the system.

7. BUDGET AND JUSTIFICATION

The proposed budget covers the resources required to design, develop, train, evaluate, and demonstrate the **Intelligent Tax Advisory Chatbot Agent** within the planned research period. Since this component is primarily based on **software development, natural language processing models, and knowledge retrieval techniques**, the budget mainly focuses on computing resources, dataset preparation, experimentation support, and prototype demonstration rather than extensive hardware procurement.

Each budget item directly supports the successful implementation and evaluation of the chatbot system, including **model experimentation, tax knowledge base preparation, conversational AI development, and system validation**.

7.1 Estimated Budget

Table 6 – Estimated Budget for the Component

Resource Type	Purpose	Estimated Amount (LKR)
Cloud / GPU compute credits	Model training, experimentation with NLP models, and evaluation of chatbot performance	40,000
Data storage and backup	Secure storage of tax knowledge datasets, model outputs, conversation logs, and experiment results	8,000
Internet and online research support	Access to research papers, datasets, model libraries, and cloud-based development tools	12,000

Resource Type	Purpose	Estimated Amount (LKR)
Software / API / productivity services	Development support tools, NLP APIs, collaboration software, and prototype integration services	10,000
Knowledge base preparation and dataset creation	Preparing tax policy datasets, regulatory document summaries, and annotation support for knowledge retrieval experiments	15,000
Testing and prototype demonstration expenses	Preparing chatbot demo environments, visualization materials, evaluation reports, and presentation-ready prototypes	10,000
Contingency allocation	Unexpected research expenses such as additional compute usage or extended experimentation	10,000

Total Estimated Budget

105,000 LKR

7.2 Budget Justification

Cloud / GPU compute credits

This allocation supports experimentation with natural language processing models used for **query understanding, intent detection, and conversational response generation**. Since transformer-based models and embedding-based retrieval techniques require significant computational resources, cloud-based GPU environments provide a cost-effective alternative to purchasing dedicated hardware.

Data storage and backup

The chatbot system will generate various research artifacts, including **tax knowledge datasets, trained models, experiment logs, and evaluation results**. Secure storage and backup solutions are required to ensure the integrity and availability of these resources during the development process.

Internet and online research support

The research process relies heavily on **access to academic publications, pretrained models, open-source NLP libraries, and tax policy datasets**. Reliable internet connectivity is therefore necessary for downloading research materials and maintaining access to development tools.

Software / API / productivity services

While the majority of development can be carried out using open-source technologies, certain productivity or experimentation tools may require limited paid subscriptions. These may include **NLP APIs, model experimentation platforms, collaboration tools, or chatbot development frameworks** that help streamline the research workflow.

Knowledge base preparation and dataset creation

One of the core challenges in developing the chatbot system is constructing a **structured tax knowledge base** that contains relevant regulatory information. This allocation supports tasks such as **collecting tax policy documents, summarizing regulatory content, organizing knowledge entries, and preparing datasets for evaluation**.

Testing and prototype demonstration expenses

This cost supports the preparation of **system evaluation materials, chatbot interface demonstrations, performance visualization outputs, and presentation-ready prototypes**. These artifacts are essential for demonstrating the effectiveness of the proposed system during research evaluation.

Contingency allocation

A contingency allocation is included to accommodate unexpected research-related expenses, such as additional compute requirements, extended experimentation periods, or unforeseen development needs.

7.3 Budget Feasibility

The proposed budget is moderate and feasible for a **software-centered research component**. Unlike projects requiring specialized hardware, sensors, or embedded systems, this component primarily relies on **computing resources, NLP models, and structured knowledge datasets**.

As a result, the overall cost remains relatively low while still being sufficient to support **model development, experimentation, system integration, and prototype demonstration**.

7.4 Cost Optimization Considerations

To minimize unnecessary expenses, the project will prioritize the use of:

- **open-source natural language processing libraries and frameworks**,
- **free academic research tools and development environments**,
- **publicly available tax policy datasets and documentation**,
- **cloud-based computing only when local resources are insufficient**,
- and **efficient dataset preparation strategies instead of expensive large-scale manual annotation**.

This approach ensures that the research remains **cost-efficient while maintaining strong technical quality and experimental validity**.

8. WORK BREAKDOWN STRUCTURE (WBS)

Figure 3 – Work Breakdown Structure and Deliverables

WP1: Requirement Analysis and Research Planning This work package focuses on defining the research scope within the Sri Lankan tax jurisdiction. It ensures that the **Intelligent Tax Advisory Language Model** aligns with the identified "Trust Gap" and establishes a roadmap for achieving **procedural alignment** in tax calculations.

WP2: Literature Review and Research Gap Analysis This involves a critical review of 2025–2026 research regarding the **Hallucination and Traceability Gap** in legal AI. The purpose is to refine the "Black Box" problem and justify the shift toward a **Neuro-Symbolic architecture**.

WP3: Tax Knowledge Graph Construction & Lex Specialis Mapping Instead of a flat knowledge base, this package focuses on building a **Knowledge Graph** in Neo4j or RDF. It includes mapping the relationships between the **Inland Revenue Act No. 24 of 2017** and the **2025 amendments**, implementing **Lex Specialis tagging** to ensure legal priority during retrieval.

WP4: Domain-Specific SLM Fine-Tuning & NLU Development This involves developing the **Small Language Model (SLM)** pipeline. The model is fine-tuned on Sri Lankan tax syntax and "Singlish" query patterns to ensure high-accuracy intent detection and entity extraction.

WP5: GraphRAG Retrieval & Priority Logic Development This work package focuses on implementing the **Graph-Augmented Retrieval (GraphRAG)** mechanism. It ensures that user queries trigger a relationship-based traversal of the Knowledge Graph, retrieving the most recent and specific laws as mandated by the legal hierarchy.

WP6: Agentic "Think Twice" Loop & Symbolic Rule Integration This stage develops the **Adversarial Agentic loop**. A "Critic Agent" is built to extract advice from the SLM and validate it against a **Symbolic Rule Engine** containing hard-coded tax brackets, effectively eliminating hallucinations.

WP7: Traceability & Proof Map Visualization Focuses on the transparency layer by developing the **Proof Map visualizer**. This module generates an auditable "paper trail" that shows the logical path from the legal source to the final advisory response.

WP8: Evaluation and Experimental Validation This involves testing the system against baseline LLMs. Evaluation metrics focus on the **Hallucination Rate**, **Legal Consistency Score**, and **Traceability Accuracy** using a mock dataset of Sri Lankan tax scenarios.

WP9: Prototype Integration and API Development Integrates the Neuro-Symbolic pipeline into a functional chatbot prototype. It includes developing the backend API (FastAPI) and a frontend interface capable of displaying **Visual Proof Maps**.

WP10: Documentation and Final Thesis Reporting The final package includes preparing the technical documentation, completing the research thesis, and finalizing the presentation materials for the submission deadline.

9. GANTT CHART

The project follows a **12-month research plan** (M1 – M12) as outlined in the system timeline. The Gantt chart provides a visual representation of the concurrent and sequential tasks required to move from data preparation to system integration.

- **M1 – M2 (Planning & Review):** Focused on **WP1** and **WP2**. Defining the problem statement and analyzing recent 2025 legal AI papers.
- **M3 – M4 (Data & Graph Engineering):** Execution of **WP3**. Constructing the Tax Knowledge Graph and gathering synthetic "Singlish" query datasets.
- **M5 – M7 (Core Development):** Intensive development of **WP4 (SLM)**, **WP5 (GraphRAG)**, and **WP6 (Agentic Loop)**. This is the primary coding phase for the Neuro-Symbolic reasoning engine.
- **M8 – M9 (Traceability & Validation):** Implementation of **WP7 (Proof Maps)** followed by the experimental testing in **WP8** to measure hallucination reduction.
- **M10 – M12 (Integration & Finalization):** Final integration (**WP9**) and the completion of the thesis documentation (**WP10**) for the March 2026 deadline.

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LIST OF APPENDICES

Appendix A – High-Level Architecture of the Intelligent Tax Advisory Chatbot Component

Appendix B – Work Breakdown Structure Diagram

Appendix C – Chatbot System Interaction Flow